

Data Center Cooling/AI Water Usage

Posted by u/mcfearless0214 • 0 points • 46 comments

My understanding was the the cooling system used in AI data centers was a closed loop system. That they weren't siphoning off water every time they needed to train their models or execute a command. However, I can't really find any articles or literature that explain how these cooling systems work in any detail but you find a lot about how AI is gonna take all of our water. Anyone have any links that explain these systems? I'd like to be able counter some of the more sensationalist talking points but I wanna make sure that I'm actually sharing correct information when I do so.

Comments

u/Rude-Distribution784 • 1 points • 2025-12-02 23:36 UTC

My Pc has a closed custom loop why dont they?

u/Sinicide7 • 1 points • 2025-11-04 11:05 UTC

I worked at Sprint before the merge and there data centers were cooled with around 10 huge Liebert DH units.

Microsoft uses huge Vertiv evaporative systems using water. They would run mostly well water but the scale on the media would require changing out a lot. They also would pull from city when demand needed it.

I also heard rumors that they built an offshore, underwater data centers were cooled with sea water in a loop. Hope this helps.

u/atlasline • 1 points • 2025-01-21 05:30 UTC

MIT white paper: The Climate and Sustainability Implications of Generative AI
<https://mit-genai.pubpub.org/pub/8ulgrckc/release/2>

u/Human_certified • 1 points • 2025-01-09 23:06 UTC

The problem is that you're arguing against mostly misunderstandings and exaggerations, or even completely baseless claims. These people don't have factual sources of their own, but they "know" this, because "everyone says".

There is plenty of information on data center water cooling, but it's mostly written by experts for experts and IT decision-makers. None of it is going to explicitly say that: "By the way, some say \[wildly exaggerated claim\], but here's why this is not true." Because nobody in the business actually thinks that in the first place.

There was some minor hype about fully submersible data centers a few years ago. Potentially very efficient in terms of cooling power usage, but I'm guessing that would now be reframed as: "OMG AI bros have invented a LITERAL IMMERSION HEATER for the PLANET!"

u/EngineerBig1851 • 0 points • 2025-01-09 22:38 UTC

If you'd rather believe populist articles cannibalising on each other rather than google "datacenter cooling methods" - I don't think a direct link to anything will crack through.

u/mcfearless0214 • 2 points • 2025-01-09 22:42 UTC

I didn't say I believed them. I came here asking for sources to counter them.

u/dreambotter42069 • 2 points • 2025-01-09 22:34 UTC

Water usage to power ChatGPT is kinda complicated.. but basically yea it needs water and we're screwed if we fuck around too much with open-loop systems. Water impacts also applies to entire nuclear power plants that turn riverwater into heated/chemically treated effluent, like Three Mile Island which is going to be restarted solely for Microsoft's AI (projected to start in 2028)

<https://abcnews.go.com/US/mile-island-site-1979-nuclear-reactor-accident-reopening/story?id=113870404>

Microsoft in 2023: "we use direct outside air most of the year to cool servers. We otherwise cool through direct evaporation" - <https://query.prod.cms.rt.microsoft.com/cms/api/am/binary/RW15mgm>

Also in 2022 when GPT-4 finished training, the West Des Moines Water Works basically said "this is too much water homie, we aint finna build more datacenters here if you don't chill"

<https://apnews.com/article/chatgpt-gpt4-iowa-ai-water-consumption-microsoft-f551fde98083d17a7e8d904f8be822c4>

<https://www.datacenterfrontier.com/machine-learning/article/33015123/microsoft-unveils-custom-designed-data-center-ai-chips-racks-and-liquid-cooling>